

Vihaan Gupta

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[GitHub](#) — [LinkedIn](#) — [Portfolio](#)

Summary

Robotics software developer focused on manipulation and perception systems, with hands-on experience in ROS2 and computer vision. Currently pursuing B.Tech in Computer Science at IIITDM Kancheepuram and actively contributing to competitive robotics projects.

Skills

Languages: C++, Python
Robotics & Simulation: ROS2, Gazebo, RViz, Isaac Sim, MoveIt2
Perception: OpenCV, YOLO, Image Processing
Core CS: Data Structures & Algorithms, OOP

Experience & Projects

- Implemented a **Cartesian control pipeline** for a 5-DOF manipulator using MoveIt2, including hardware interface, microcontroller communication, and trajectory execution.
- Developed **perception pipelines using camera and LiDAR data**, including depth-based processing for extracting positions of objects identified using YOLO models.
- Trained and deployed a **YOLOv8-based object detection model** for outdoor cone detection in autonomous rover tasks.
- Worked on **system integration and real-time debugging** across perception, control, and navigation subsystems during field testing.
- Collaborated with the navigation team working on **autonomous driving pipeline development**.
- Currently developing a **full-stack autonomous manipulation system** for pick-and-place in unknown environments, focusing on object detection, pose estimation, and grasp planning.

Competitions & Achievements

- Secured **10th rank globally** in International Rover Challenge (IRC), 2026.
- Awarded **Best Innovation Award** at Caterpillar Autonomy Challenge, IIT Madras.
- Ranked in **top 25 nationwide** in 2 innovation challenges at Techfest IIT Bombay.
- Currently participating in European Rover Challenge (ERC).

Education

B.Tech in Computer Science and Engineering IIITDM Kancheepuram	2024 – Present
Class XII (PCM) St. Joseph's Co-ed School, Bhopal	2024

Languages

English, Hindi